From Paper to Digital

Gaurav Sood

When we think about paper . . .

We think about government offices






circle of blue

But paper based storage of information is common

Libraries and Archives

Health records

Receipts . . .

Small Businesses

And it isn't going away (quickly).

The Dead Tree Format

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  Digital indexes can make text searchable.
- Hard to analyze, summarize stored information
- Hard to track performance, identify anomalous transactions, identify patterns ...

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  Evaluate a representative sample of your documents before choosing an OCR workflow.

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  - Abbyy FineReader
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- But ...
  - Accuracy depends on layout, language, scan quality, and the recognition model.
  - Confidence scores are not measured accuracy: validate against transcribed examples.
  - Scalability depends on throughput, hardware, and review requirements.

How to Convert Squiggles to Bits?

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– Take images of paper

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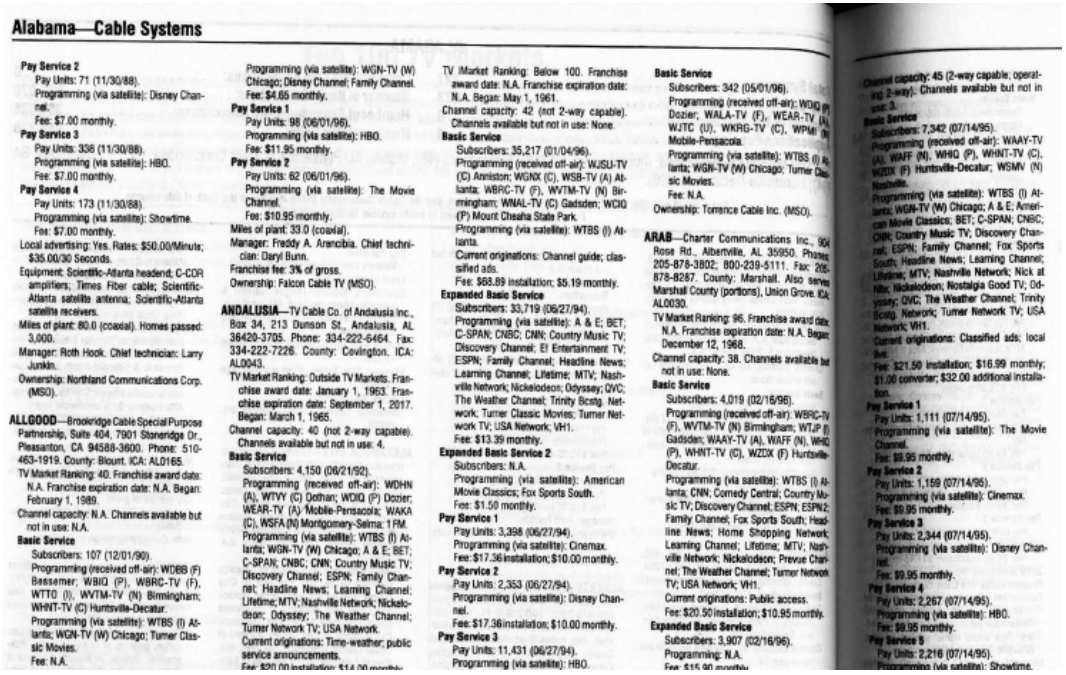
- Take images of paper
- Within images, find where relevant text is located
- Find out how the text is laid out
- Recognize the characters

Thus Performance Depends on...

– Quality of the scan: spine, contrast etc.

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Alabama—Cable Systems

Pay Service 2
Pay Units: 71 (11/30/88).
Programming (via satellite): Disney Channel.
Fee: $7.00 monthly.
Pay Service 3
Pay Units: 338 (11/30/88).
Programming (via satellite): HBO.
Fee: $7.00 monthly.
Pay Service 4
Pay Units: 173 (11/30/88).
Programming (via satellite): Showtime.
Fee: $7.00 monthly.
Local advertising: Yes. Rates: $50.00/Minute; $35.00/30 Seconds.
Equipment: Scientific-Atlanta headend; C-COR amplifiers; Times Fiber cable; Scientific-Atlanta satellite antenna; Scientific-Atlanta satellite receivers.
Miles of plant: 80.0 (coaxial). Homes passed: 3,000.
Manager: Roth Hook. Chief technician: Larry Junkin.
Ownership: Northland Communications Corp. (MSO).

ALLGOOD—Brookridge Cable Special Purpose Partnership, Suite 404, 7901 Stoneridge Dr., Pleasanton, CA 94588-3600. Phone: 510-463-1919. County: Blount. ICA: AL0165.
TV Market Ranking: 40. Franchise award date: N.A. Franchise expiration date: N.A. Began: February 1, 1989.
Channel capacity: N.A. Channels available but not in use: N.A.
Basic Service
Subscribers: 107 (12/01/90).
Programming (received off-air): WDBB (F) Bessemer; WBIQ (P), WBRC-TV (F), WTTO (I), WVTM-TV (N) Birmingham; WHNT-TV (C) Huntsville-Decatur.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; Turner Classic Movies.
Fee: N.A.
Programming (via satellite): WGN-TV (W) Chicago; Disney Channel; Family Channel.
Fee: $4.65 monthly.
Pay Service 1
Pay Units: 98 (06/01/96).
Programming (via satellite): HBO.
Fee: $11.95 monthly.
Pay Service 2
Pay Units: 62 (06/01/96).
Programming (via satellite): The Movie Channel.
Fee: $10.95 monthly.
Miles of plant: 33.0 (coaxial).
Manager: Freddy A. Arancibia. Chief technician: Daryl Bunn.
Franchise fee: 3% of gross.
Ownership: Falcon Cable TV (MSO).

ANDALUSIA—TV Cable Co. of Andalusia Inc., Box 34, 213 Dunson St., Andalusia, AL 36420-3705. Phone: 334-222-6464. Fax: 334-222-7226. County: Covington. ICA: AL0043.
TV Market Ranking: Outside TV Markets. Franchise award date: January 1, 1963. Franchise expiration date: September 1, 2017. Began: March 1, 1965.
Channel capacity: 40 (not 2-way capable). Channels available but not in use: 4.
Basic Service
Subscribers: 4,150 (06/21/92).
Programming (received off-air): WDHN (A), WTVY (C) Dothan; WDIQ (P) Dozier; WEAR-TV (A) Mobile-Pensacola; WAKA (C), WSFA (N) Montgomery-Selma; 1 FM.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; A & E; BET; C-SPAN; CNBC; CNN; Country Music TV; Discovery Channel; ESPN; Family Channel; Headline News; Learning Channel; Lifetime; MTV; Nashville Network; Nickelodeon; Odyssey; The Weather Channel; Turner Network TV; USA Network.
Current originations: Time-weather; public service announcements.
Fee: $20.00 installation; $14.00 monthly.
TV Market Ranking: Below 100. Franchise award date: N.A. Franchise expiration date: N.A. Began: May 1, 1961.
Channel capacity: 42 (not 2-way capable). Channels available but not in use: None.
Basic Service
Subscribers: 35,217 (01/04/96).
Programming (received off-air): WJSU-TV (C) Anniston; WGNX (C), WSB-TV (A) Atlanta; WBRC-TV (F), WVTM-TV (N) Birmingham; WNAL-TV (C) Gadsden; WCIQ (P) Mount Cheaha State Park.
Programming (via satellite): WTBS (I) Atlanta.
Current originations: Channel guide; classified ads.
Fee: $68.89 installation; $5.19 monthly.
Expanded Basic Service
Subscribers: 33,719 (06/27/94).
Programming (via satellite): A & E; BET; C-SPAN; CNBC; CNN; Country Music TV; Discovery Channel; El Entertainment TV; ESPN; Family Channel; Headline News; Learning Channel; Lifetime; MTV; Nashville Network; Nickelodeon; Odyssey; QVC; The Weather Channel; Trinity Bcstg. Network, Turner Classic Movies; Turner Network TV; USA Network; VH1.
Fee: $13.39 monthly.
Expanded Basic Service 2
Subscribers: N.A.
Programming (via satellite): American Movie Classics; Fox Sports South.
Fee: $1.50 monthly.
Pay Service 1
Pay Units: 3,398 (06/27/94).
Programming (via satellite): Cinemax.
Fee: $17.36 installation; $10.00 monthly.
Pay Service 2
Pay Units: 2,353 (06/27/94).
Programming (via satellite): Disney Channel.
Fee: $17.36 installation; $10.00 monthly.
Pay Service 3
Pay Units: 11,431 (06/27/94).
Programming (via satellite): HBO.
Basic Service
Subscribers: 342 (05/01/96).
Programming (received off-air): WDIQ (P) Dozier; WALA-TV (F), WEAR-TV (A) Mobile-Pensacola; WJTC (U), WKRG-TV (C), WPMI (N) Mobile-Pensacola.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; Turner Classic Movies.
Fee: N.A.
Ownership: Torrence Cable Inc. (MSO).

ARAB—Charter Communications Inc., 904 Rose Rd., Albertville, AL 35950. Phone: 205-878-3802; 800-239-5111. Fax: 205-878-8287. County: Marshall. Also serves Marshall County (portions), Union Grove. ICA: AL0030.
TV Market Ranking: 96. Franchise award date: N.A. Franchise expiration date: N.A. Began: December 12, 1968.
Channel capacity: 38. Channels available but not in use: None.
Basic Service
Subscribers: 4,019 (02/16/96).
Programming (received off-air): WBRC-TV (F), WVTM-TV (N) Birmingham; WTJP (I) Gadsden; WAAY-TV (A), WAFF (N), WHIQ (P), WHNT-TV (C), WZDX (F) Huntsville-Decatur.
Programming (via satellite): WTBS (I) Atlanta; CNN; Comedy Central; Country Music TV; Discovery Channel; ESPN; ESPN2; Family Channel; Fox Sports South; Headline News; Home Shopping Network; Learning Channel; Lifetime; MTV; Nashville Network; Nickelodeon; Prevue Channel; The Weather Channel; Turner Network TV; USA Network; VH1.
Current originations: Public access.
Fee: $20.50 installation; $10.95 monthly.
Expanded Basic Service
Subscribers: 3,907 (02/16/96).
Programming: N.A.
Fee: $15.90 monthly.

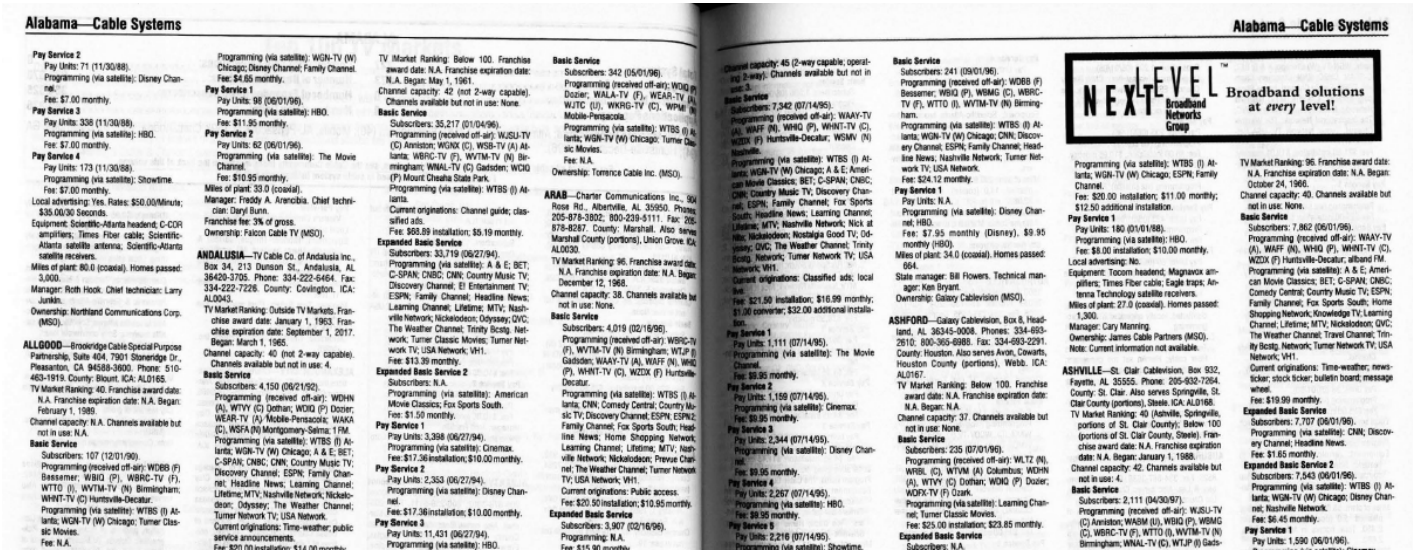
Channel capacity: 45 (2-way capable; operating 2-way): Channels available but not in use: 3.
Basic Service
Subscribers: 7,342 (07/14/95).
Programming (received off-air): WAAY-TV (A), WAFF (N), WHIQ (P), WHNT-TV (C), WZDX (F) Huntsville-Decatur; WSMV (N) Nashville.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; A & E; American Movie Classics; BET; C-SPAN; CNBC; CNN; Country Music TV; Discovery Channel; ESPN; Family Channel; Fox Sports South; Headline News; Learning Channel; Lifetime; MTV; Nashville Network; Nick at Nite; Nickelodeon; Nostalgia Good TV; Odyssey; QVC; The Weather Channel; Trinity Bcstg. Network; Turner Network TV; USA Network; VH1.
Current originations: Classified ads; local live.
Fee: $21.50 installation; $16.99 monthly; $1.00 converter; $32.00 additional installation.
Pay Service 1
Pay Units: 1,111 (07/14/95).
Programming (via satellite): The Movie Channel.
Fee: $9.95 monthly.
Pay Service 2
Pay Units: 1,159 (07/14/95).
Programming (via satellite): Cinemax.
Fee: $9.95 monthly.
Pay Service 3
Pay Units: 2,344 (07/14/95).
Programming (via satellite): Disney Channel.
Fee: $9.95 monthly.
Pay Service 4
Pay Units: 2,267 (07/14/95).
Programming (via satellite): HBO.
Fee: $9.95 monthly.
Pay Service 5
Pay Units: 2,216 (07/14/95).
Programming (via satellite): Showtime.

Thus Performance Depends on...

- Quality of the scan: spine, contrast etc.
- Complexity of the layout

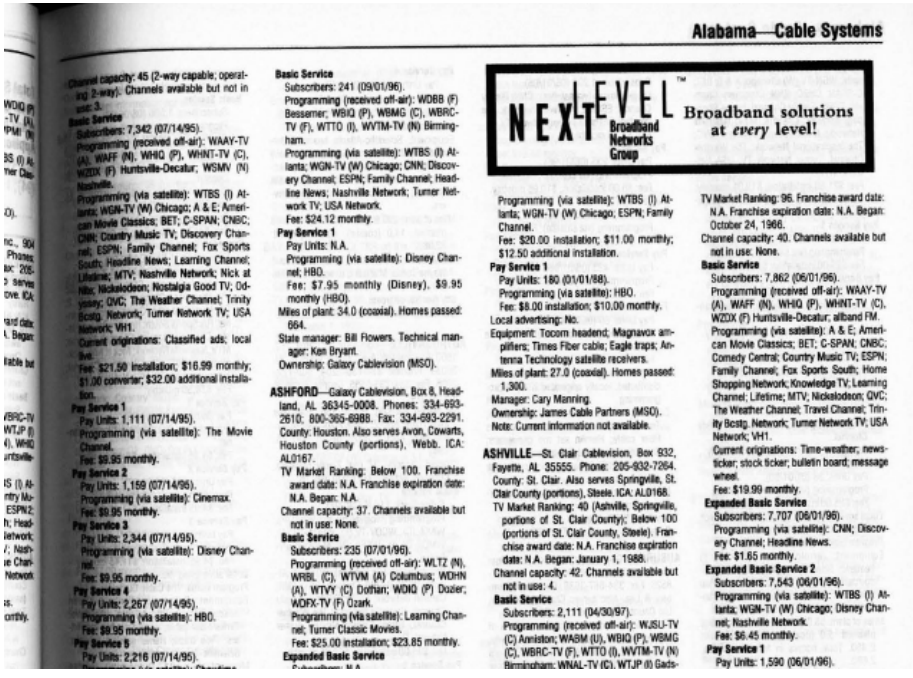
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Programming (received off-air): WAAY-TV (A), WAFF (N), WHIQ (P), WHNT-TV (C), WZDX (F) Huntsville-Decatur; WSMV (N) Nashville.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; A & E; American Movie Classics; BET; C-SPAN; CNBC; CNN; Country Music TV; Discovery Channel; ESPN; Family Channel; Fox Sports South; Headline News; Learning Channel; Lifetime; MTV; Nashville Network; Nick at Nite; Nickelodeon; Nostalgia Good TV; Odyssey; QVC; The Weather Channel; Trinity Bcstg. Network; Turner Network TV; USA Network; VH1.
Current originations: Classified ads; local live.
Fee: $21.50 installation; $16.99 monthly; $1.00 converter; $32.00 additional installation.
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Pay Service 4
Pay Units: 2,267 (07/14/95).
Programming (via satellite): HBO.
Fee: $9.95 monthly.
Pay Service 5
Pay Units: 2,216 (07/14/95).

Basic Service
Subscribers: 241 (09/01/96).
Programming (received off-air): WDBB (F) Bessemer; WBIQ (P), WBMG (C), WBRC-TV (F), WTTO (I), WVTM-TV (N) Birmingham.
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; CNN; Discovery Channel; ESPN; Family Channel; Headline News; Nashville Network; Turner Network TV; USA Network.
Fee: $24.12 monthly.
Pay Service 1
Pay Units: N.A.
Programming (via satellite): Disney Channel; HBO.
Fee: $7.95 monthly (Disney), $9.95 monthly (HBO).
Miles of plant: 34.0 (coaxial). Homes passed: 664.
State manager: Bill Flowers. Technical manager: Ken Bryant.
Ownership: Galaxy Cablevision (MSO).

ASHFORD—Galaxy Cablevision, Box 8, Headland, AL 36345-0008. Phones: 334-693-2610; 800-365-6988. Fax: 334-693-2291. County: Houston. Also serves Avon, Cowarts, Houston County (portions), Webb. ICA: AL0167.
TV Market Ranking: Below 100. Franchise award date: N.A. Franchise expiration date: N.A. Began: N.A.
Channel capacity: 37. Channels available but not in use: None.
Basic Service
Subscribers: 235 (07/01/96).
Programming (received off-air): WLTZ (N), WRBL (C), WTVM (A) Columbus; WDHN (A), WTVY (C) Dothan; WDIQ (P) Dozier; WDFX-TV (F) Ozark.
Programming (via satellite): Learning Channel; Turner Classic Movies.
Fee: $25.00 installation; $23.85 monthly.
Expanded Basic Service

NEXTLEVEL™ Broadband Networks Group
Broadband solutions at *every* level!

Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; ESPN; Family Channel.
Fee: $20.00 installation; $11.00 monthly; $12.50 additional installation.
Pay Service 1
Pay Units: 180 (01/01/88).
Programming (via satellite): HBO.
Fee: $8.00 installation; $10.00 monthly.
Local advertising: No.
Equipment: Tocom headend; Magnavox amplifiers; Times Fiber cable; Eagle traps; Antenna Technology satellite receivers.
Miles of plant: 27.0 (coaxial). Homes passed: 1,300.
Manager: Cary Manning.
Ownership: James Cable Partners (MSO).
Note: Current information not available.

ASHVILLE—St. Clair Cablevision, Box 932, Fayette, AL 35555. Phone: 205-932-7264. County: St. Clair. Also serves Springville, St. Clair County (portions), Steele. ICA: AL0168.
TV Market Ranking: 40 (Ashville, Springville, portions of St. Clair County); Below 100 (portions of St. Clair County, Steele). Franchise award date: N.A. Franchise expiration date: N.A. Began: January 1, 1988.
Channel capacity: 42. Channels available but not in use: 4.
Basic Service
Subscribers: 2,111 (04/30/97).
Programming (received off-air): WJSU-TV (C) Anniston; WABM (U), WBIQ (P), WBMG (C), WBRC-TV (F), WTTO (I), WVTM-TV (N) Birmingham; WNAL-TV (C), WTJP (I) Gads-

TV Market Ranking: 96. Franchise award date: N.A. Franchise expiration date: N.A. Began: October 24, 1966.
Channel capacity: 40. Channels available but not in use: None.
Basic Service
Subscribers: 7,862 (06/01/96).
Programming (received off-air): WAAY-TV (A), WAFF (N), WHIQ (P), WHNT-TV (C), WZDX (F) Huntsville-Decatur; allband FM.
Programming (via satellite): A & E; American Movie Classics; BET; C-SPAN; CNBC; Comedy Central; Country Music TV; ESPN; Family Channel; Fox Sports South; Home Shopping Network; Knowledge TV; Learning Channel; Lifetime; MTV; Nickelodeon; QVC; The Weather Channel; Travel Channel; Trinity Bcstg. Network; Turner Network TV; USA Network; VH1.
Current originations: Time-weather; newsticker; stock ticker; bulletin board; message wheel.
Fee: $19.99 monthly.
Expanded Basic Service
Subscribers: 7,707 (06/01/96).
Programming (via satellite): CNN; Discovery Channel; Headline News.
Fee: $1.65 monthly.
Expanded Basic Service 2
Subscribers: 7,543 (06/01/96).
Programming (via satellite): WTBS (I) Atlanta; WGN-TV (W) Chicago; Disney Channel; Nashville Network.
Fee: $6.45 monthly.
Pay Service 1
Pay Units: 1,590 (06/01/96).

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- Complexity of the layout
- Font
- Language
- Hardware and Software (duh!)

A Character-Based OCR Pipeline

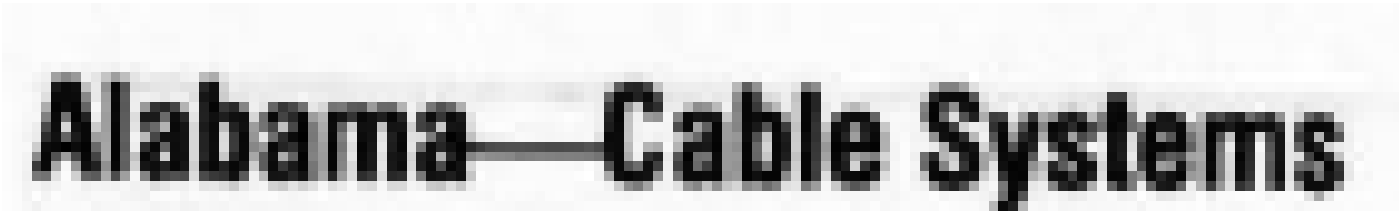
– Make images

A Character-Based OCR Pipeline

- Make images
- Detect Text

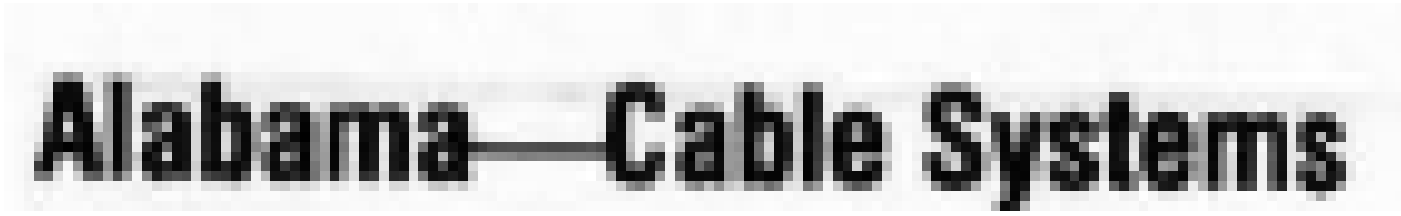
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- Segment “Characters”

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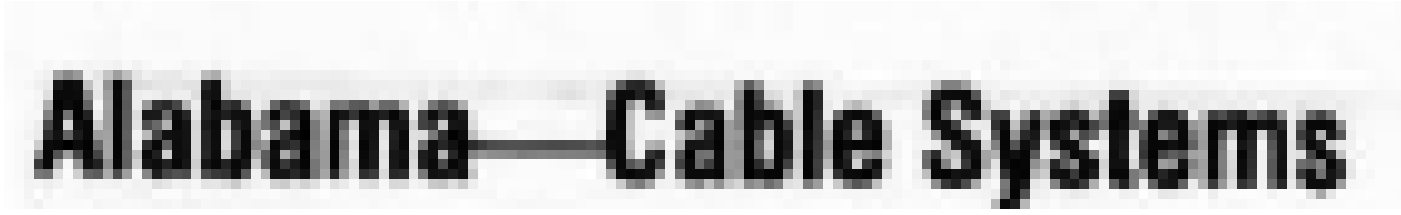
- Segment "Characters"



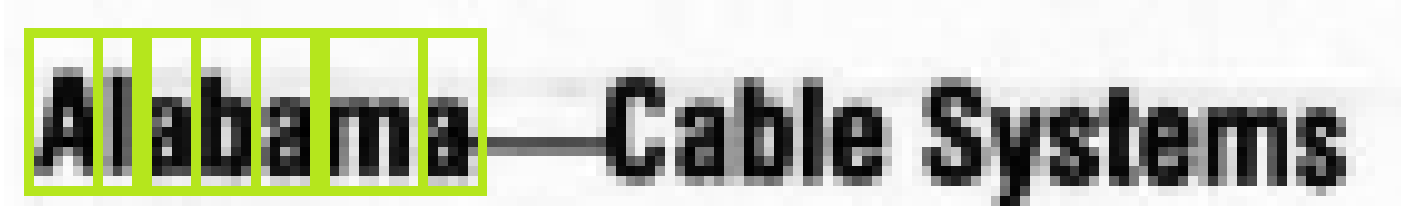
- Classify "Characters"

A Character-Based OCR Pipeline

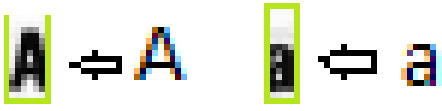
- Make images
- Detect Text



- Segment "Characters"



- Classify "Characters"



Recognizing Text Sequences

- Character segmentation is one approach, not a requirement of every OCR system.
- Sequence models can recognize whole text lines without first assigning a box to each character.
- Language support depends on the trained model. Tesseract supports many languages and scripts.

Tesseract documentation

Character-Based Recognition

- Detect Text

Character-Based Recognition

- Detect Text
  - Supervised Learning

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  - Blobs with text, Blobs without

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  - Letters (and Ligatures) versus Splits

Character-Based Recognition

- Detect Text
  - Supervised Learning
  - Blobs with text, Blobs without
  - But size of a blob is an issue
- Character Segmentation
  - Supervised Learning
  - Letters (and Ligatures) versus Splits
- Classify Characters (and Ligatures)
  - Supervised Learning
  - A versus B versus C...

Supervised Learning

– Classified (training) data

Supervised Learning

- Classified (training) data
- Estimate a model

Supervised Learning

- Classified (training) data
- Estimate a model
  $logit[p(spam)] = \alpha + f'\beta$ where f is frequencies.

Supervised Learning

- Classified (training) data
- Estimate a model
  Predict class (e.g. Blobs with or without text) using features (pixel by pixel rgb)
  Use cross-validation to tune the parameters

Supervised Learning

- Classified (training) data
- Estimate a model
- Predict classes of unseen data (groups of pixels)

Paper to Digital Pipeline

- Take images of paper
- Within images, find where relevant text is located
- Find out how the text is laid out
- Recognize the characters

Paper to Digital Pipeline

- Take images of paper
- Within images, find where relevant text is located
- Find out how the text is laid out
- Recognize the characters
- Every step is error prone

Optimize all steps w.r.t final error rate.

Optimize all steps w.r.t final error rate. How to deal with errors that remain

How to Fix Errors

– How confident are you that...

How to Fix Errors

- How confident are you that...
  - An area has relevant text

How to Fix Errors

- How confident are you that...
  - An area has relevant text
  - Split is correct

How to Fix Errors

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  - Right character (or ligature) is recognized

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  - An area has relevant text
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  - Right character (or ligature) is recognized
- Flag low confidence areas, splits, characters...

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How to Fix Errors

- How confident are you that...
  - An area has relevant text
  - Split is correct
  - Right character (or ligature) is recognized
- Flag low confidence areas, splits, characters...
- Get humans to identify the correct classes
- Use that knowledge to fix other errors

Fixing Character Recognition Errors

– Search and Replace

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- Search and Replace
- OCR makes certain kinds of errors (| is mistaken for an I)

Fixing Character Recognition Errors

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Fixing Character Recognition Errors

- Search and Replace
- OCR makes certain kinds of errors (| is mistaken for an I)
- Compare against a corpus (dictionary) and replace
- But replace with what?
- standd -> strand, stand, stood, or sand?

Edit Distance

– How similar are two strings?

Edit Distance

- How similar are two strings?
- Typically refers to minimum edit distance

Edit Distance

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- Measure character and word error rates on a separate, randomly sampled, hand-transcribed test set.